

Wallfacer Ultimate Configuration — One Step to Completion

Core Philosophy

Do not buy enterprise-grade (H100, Xeon Platinum) — that is for 100-person teams. Buy top consumer/workstation-grade hardware; sufficient for a single person.

Ultimate Configuration Spec

Component	Model	Qty	Unit Price	Amount	Rationale
CPU	AMD Threadripper PRO 7995WX (96C/192T)	1	¥85,000	¥85,000	Fastest x86 globally. Molecular finger-prints all-core parallel, 12 databases parsed simultaneously
GPU	NVIDIA RTX 5090 32GB	4	¥18,000	¥72,000	4-way parallel Yajie + NEP training + CC inference. 5090 is 70% faster than 4090

Component	Model	Qty	Unit Price	Amount	Rationale
Memory	512GB DDR5- 5600 ECC (8×64GB)	1 set	¥16,000	¥16,000	All databases + interme- diate results in-memory mapped. Zero disk waiting
Motherboard	ASUS Pro WS WRX90E- SAGE SE	1	¥12,000	¥12,000	7× PCIe 5.0 x16, 4 GPU + high-speed network
System Disk	Samsung 990 Pro 4TB NVMe PCIe 4.0	2 (RAID1)	¥2,500	¥5,000	OS + hot data, mirrored redun- dancy
Data Disk	Samsung 990 Pro 4TB NVMe PCIe 4.0	4 (RAID0)	¥2,500	¥10,000	16TB high-speed array, 28GB/s read/write
Backup Disk	WD Gold 24TB HDD	2 (RAID1)	¥4,000	¥8,000	24TB redundant cold backup
PSU	Corsair AX1600i 1600W Titanium	2 (redun- dant)	¥3,500	¥7,000	4 GPU full load ~1400W, dual PSU redun- dancy

Component	Model	Qty	Unit Price	Amount	Rationale
Cooling	EK-Quantum custom water cooling (CPU + 4 GPU)	1 set	¥20,000	¥20,000	7995WX full load 350W, 4×5090 full load 1800W, must water cool
Case	Corsair 1000D Super Full-Tower	1	¥3,000	¥3,000	Dual 480mm + dual 360mm radiator positions
UPS	APC SRT3000XLI 3000VA	1	¥8,000	¥8,000	15 min runtime at full load, safe shutdown
Monitor	Dell U4323QE 43" 4K IPS	1	¥6,000	¥6,000	Large screen: papers + code + logs side by side
Network	Mellanox ConnectX-6 100GbE	1	¥3,000	¥3,000	Supercomputer direct 100Gbps link
Keyboard	HHKB Professional HYBRID Type-S	1	¥2,000	¥2,000	A wallfacer deserves a good keyboard

Total Cost

Category	Amount
Core Compute (CPU+GPU+Motherboard)	¥169,000
Storage (NVMe+HDD)	¥23,000
Cooling+Case+PSU	¥30,000
Peripherals+Network+UPS	¥19,000
Memory	¥16,000
Total	¥257,000

Performance Estimates

Task	Plan B (¥45K)	Ultimate (¥257K)	Speedup
Molecular fingerprints (10^7 molecules)	2 hours	15 min	8×
Yajie Full Screening (10^8 pairs)	8-12 hours	1.5 hours	6×
CC parallel math derivation	3 ways	8 ways	2.7×
NEP training (AIN 534 frames)	1 GPU	4 GPU data parallel	3.5×
Load 200GB DB into memory	Disk I/O bound	Seconds	(all in mem)

What This Machine Can Do

- **12 Database Full Screening:** Start at 2 PM, results by 4 PM
- **8-way CC parallel hostile review:** Paper written, immediately attacked by 8 reviewers simultaneously
- **NEP training + Yajie audit run concurrently:** No resource contention
- **All data in memory:** 200GB database arbitrary joins, no index optimization needed
- **100Gbps direct supercomputer link:** DFT data transferred in seconds

What Not to Buy

Item	Rationale
H100 80GB $\times$ 8	¥2M. 5090 $\times$ 4 is sufficient. Large VRAM useful for large model training, SCX uses M small-expert inference
NVIDIA DGX	¥3M. Enterprise pricing. Not worth it for one person
Rack/UPS server room	This is not a datacenter. One tower workstation suffices
Dual Xeon	7995WX single-socket 96 cores already exceeds dual Xeon 64 cores